

Donghwan Rho

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Research Interests

- Implementing Language Models Under Homomorphic Encryption
- Mathematical Reasoning in Large Language Models
- Interpretability of Language Models

Education

- Ph.D. Seoul National University**, Mathematical Sciences Mar. 2021 – Present
- Advisor: [Ernest K. Ryu](#)
 - Expected Graduation: Feb. 2026
- BS Korea University**, Mathematics Education Mar. 2015 – Feb. 2021
- Graduated 1st in Class
 - (Mandatory) Military Service: May 2017 - Jan. 2019
 - Teacher's Certificate for Secondary Regular Teacher of Mathematics

Experience

- Workshop Coordinator**, SNU-KAIST ML Theory Workshop Gangneung, South Korea
Aug. 2024
- Assisted in planning the schedule, selecting the venue, and managing workshop programs.
- Teaching Practicum** Seoul, South Korea
Apr. 2020 - May 2020
- Danggok high school
- Military Service**, Auxiliary Police Seoul, South Korea
May 2017 – Jan. 2019
- Assisted in maintaining public order and security.

Publications

*: Equal Contribution

- [3] **Studying the Korean Word-Chain Game with RLVR: Mitigating Reward Conflicts via Curriculum Learning** Oct. 2025
Donghwan Rho
 In progress
[Paper](#)
- [2] **Traveling Salesman-Based Token Ordering Improves Stability in Homomorphically Encrypted Language Models** Oct. 2025
Donghwan Rho*, Sieun Seo*, Hyewon Sung, Chohong Min, Ernest K. Ryu
 Submitted
[Paper](#)
- [1] **Encryption-Friendly LLM Architecture** Oct. 2024
Donghwan Rho*, Taeseong Kim*, Minje Park, Jung Woo Kim, Hyunsik Chae, Ernest K. Ryu, Jung Hee Cheon
 International Conference on Learning Representations (ICLR) 2025
[Paper](#) [Code](#)

Talks & Presentations

Seoul National University Research Institute of Mathematics Symposium , Talk	Nov. 2025
• Title: Implementing Efficient Language Models under Homomorphic Encryption	
Korean Mathematical Society Annual Meeting , Talk	Oct. 2025
• Title: Traveling Salesman-Based Token Ordering Improves Stability in Homomorphically Encrypted Language Models	
SNU-Samsung Electronics Collaboration Symposium , Poster Presentation	Aug. 2024
• Title: Encryption-Friendly LLM Architecture	
SNU-KAIST ML Theory Workshop , Poster Presentation	Aug. 2024
• Title: Encryption-Friendly LLM Architecture	

Honors and Awards

Outstanding TA , Seoul National University	July 2025
Introduction to Computer Programming and Artificial Intelligence for Scientists	
University Students Contest of Mathematics , Silver Prize	Dec. 2018
Academic Excellence Scholarship , Korea University	Aug. 2015

Teaching

Peer Tutoring in Basic Sciences , Lead Teaching Assistant	Mar. 2024 - Present
• Represented section TAs to ensure the smooth operation of the peer tutoring program.	
• Oversaw tutorings in Mathematics, Physics, Chemistry, Biology, and Statistics.	
Basic Calculus , Teaching Assistant	Sep. 2021 - Present
• Matched tutors and tutees for the <i>Basic Calculus</i> peer tutoring program.	
• Managed program logistics to ensure efficient operation.	
SNU Pre-College Program , Head TA, Instructor	
• Managed logistics of the mathematics program for freshman undergraduate.	
• Gave lectures.	
• Winter 2024, Winter 2023, Winter 2022	
Introduction to Computer Programming and Artificial Intelligence for Scientists , Instructor	Spring 2025
• English Course	
• Selected as an outstanding TA 	
Topics in Machine Intelligence	Fall 2024
• Topics: Continuous-depth neural networks, Diffusion Models, Large Language Models, Vision Language Models, State-space Models	
• English Course	
• Graduate Course	
Mathematical algorithms II	Fall 2022
• Topics: Reinforcement Learning, Diffusion Models, Natural Language Processing	
• English Course	

- Graduate Course

Calculus 2 Practice, Instructor

- Fall 2021, Fall 2025

Calculus 1 Practice, Instructor

- Spring 2024, Fall 2023, Spring 2023, Spring 2022, Spring 2021

Technical Skills

Languages: Korean (Native), English (Fluent)

Programming Languages: Python, C++

Tools & Technologies: LaTeX, GitHub